NCCA RESEARCH PROPOSAL FORM

1. Ano ang tawag sa pag-aaral na ito? *Working title of the research.*

Ethno-Medicinal Knowledge and Healthcare Practices of the Traditional Health Practitioners in the Remote East-Coastal Communities of Camarines Sur

2. Ibigay ang buod ng pag-aaral (150 salita). *Abstract of the proposed project* (150 words).

This study generally aims to document the distinct culture of remote east-coastal communities in Camarines Sur in traditional health and healthcare management. The traditional health practitioners are identified as the local experts in traditional medicine because of their notable experiences which have endured overtime and are rooted in a body of knowledge that is customarily passed on orally. Thus, collecting their practices and knowledge can preserve the nearly extinct intangible national treasures that may include unstudied herbs which can be subjected to pharmacological, botanical and/or chemical research in the quest of discovering new medicines to cure various illnesses. To materialize these, the study will utilize directed narrative content analysis to analyze the cultural concepts on illness, disease causation, diagnosis and treatment. In addition, informant consensus factor (F_{ic}), Use Value (UV), and Fidelity Levels (FL) will be used to determine the agreement among informants on the use of plants as remedies.

3. Ano ang gustong malaman sa pamamagitan ng proyektong ito? May guwang o kawalan ba ng kaalaman na mapupunuan ng proyekto? *What is the research question and what is the research gap addressed?*

This proposed study generally aims to document the cultural knowledge and healthcare practices of the traditional health practitioners in the remote coastal communities of Camarines Sur. Specifically, it aims to:

1. Describe the dynamics of traditional healthcare practitioners in terms of demographic profile, market served, origin and/or reasons for practicing folk healing, and types of folk healing practices;
2. Explore their cultural beliefs and concept of illness, disease causation, diagnosis and treatment;
3. Characterize the medicinal plants used in healing/healthcare by the traditional health practitioners in Partido Area according to:
    a. Scientific and vernacular names
    b. Taxonomic category and family
    c. Plant parts used
    d. Preparation and administration
    e. Disease/illness cured
    f. Habit, habitat and nature of the plants; and
4. Calculate the informant consensus factor (F_{ic}), Use Value (UV), and Fidelity Levels (FL) for each medicinal plant species used to cure various ailments.

This study aims to address the knowledge gaps such as:

1. There is no recorded nor published articles and the like which describes the distinct culture on health and healthcare management practices of the folk healers in Camarines Sur, these include but not limited to: concept of illness, disease causation, diagnosis and treatment; and
2. There no recorded nor published ethno-botanical knowledge or practices in Camarines Sur despite the existence of rich flora and patronization to and presence of folk healers in the province.

Thus, this calls for urgent conservation and recording for the benefit of the communities and the society, as a whole.

4. Ano-ano na ang mga pag-aaral ang naisagawa o nalathala tungkol o na may kinalaman sa bagay, gawain, lugar, atbp. Na sabjek ng riserts? Paano ang mga ito nakatutulong sa pag-aaral na ito? *Review of related literature/studies*.

Culture plays an important role in healthcare system of local communities. Every society has developed curative methods for coping demoralizing effects of disease on normal life (Jocano, 1966). Often these preventive health measures are surrounded by a substantial body of beliefs, knowledge, and practices (Scotch, 1963) which is designed to make the medicine more effective. Tan (2008) explains that culture is inscribed in one's bodies and mind. Intrinsically, it is imperative to recognize the relationship of culture and health as it influences an individuals' perspectives and decision-making process (Purnell, 2005). The said relationship can be ascribed to as the folk healing systems. Leininger (1991) in his theory of nursing, posited that all cultures in the world have had a lay health care system which is usually described as folk, indigenous or generic. Hence, in studying the culture of a local community, healthcare systems usually evolved as one of the main subjects.

The study of traditional/folk medicines and the variant medical subcultures found in various communities is called ethnomedicine. According to Hughes (1968), ethnomedicine refers to health beliefs and practices which are a result of indigenous cultural development and not explicitly derived from the conceptual framework of modern medicine. Ethnomedical studies of health intend to determine the insiders' knowledge of various systems of healing by focusing on traditional healers and alternative medicine practices (Del Fiero & Nolasco, 2013). The practices of traditional healers are considered a form of complementary and alternative medicine (National Center for Complementary and Integrative Health, 2019).

One of the common ethnomedicinal practices is the use of plants to cure one's illnesses. Accordingly, medicinal plants and herbs have been used for many centuries as a source of people's drugs for the treatment and prevention of diseases, disorders and the promotion of good health (Alpuerto, et al., 2010) and still provide the first line of primary health-care even in the present age to major segments of the population worldwide (Khatun, Or-Rashid, & Rahmatullah, 2011). According to the World Health Organization (2003), it is estimated that up to 80% of the population depends exclusively on plants for their health and healing. The information and folk knowledge regarding the medicinal and therapeutic uses of these indigenous plant materials have been handed down from generation to generation through verbal communication (Hossan, et al., 2010).

However, the nature of healthcare in the Philippines has turned towards Western medicine due to economic development, urbanization (Abe & Ohtani, 2013), and modernization (Mahmood, Mahmood, & Tabassum, 2011). The rapid disappearance of traditional culture suggests that unrecorded folk knowledge and information may be lost forever (Abe & Ohtani, 2013). Nevertheless, there are few systematic studies conducted in the Philippines which are accessible online that deals on recording these significant informational resources. These include, but not limited to, the study of Angagan, Buot, Relox and Rebancos (2010) in Apayao; Balangcod T. and Balangcod K (2015) in Benguet Province; Blasco, De Guznab and Alejandro (2014) in Surigao Del Sur; Buot (2009) in Albay; and Ong and Kim (2014) in Guimaras.

Concomitantly, there is no recorded ethno-medicinal knowledge and practices in Camarines Sur. This area has rich in diverse flora which might have played an important role in an effective ethnobotanical practice. The patronization to and presence of traditional health practitioners in the Area provides a significant evidence that there are rich ethnomedicinal knowledge. With proper documentation and scientific research, traditional medicine can solve much of basic health care service problems in remote communities (Habtom, 2018). Thus, this calls for urgent conservation and recording of these cultures for the benefit of the communities.

PAMBANSANG KOMISYON PARA SA KULTURA AT MGA SINING

January 6, 2020

MR. CALYD T. CERIO
Director
Center for Bioresource-based Enterprise Development
Partido State University – Provincial State of Agriculture
Salogon, San Jose, Camarines Sur

Dear **Mr. Cerio**:

Greetings from the National Commission for Culture and the Arts!

The NCCA is pleased to inform you that your organization's research proposal entitled **"Ethno-Medicinal Knowledge and Healthcare Practices of the Traditional Healthcare Practices in the Remote East-Coastal Communities of Camarines Sur"**, under the Individual Research Awards, was approved by the NCCA Board of Commissioners, in the amount of **Three Hundred Thousand Pesos (PhP 300,000.00)** per Board Resolution No. 2020-221.

We would appreciate your calling the Cultural Dissemination Section at telephone numbers (02) 527-2192 loc 541 and look for Ms. Angela Sonico to discuss the mechanics of your project's implementation as well as the manner of fund release, subject to availability of cash.

Very truly yours,

BERNAN JOSEPH R. CORPUZ
Chief, PPFPD

Noted by:

AL RYAN ALEJANDRE
Executive Director

5. Ilarawan o isalaysay kung paano isasagawa ang pananaliksik. Ano ang mga pangunahing ideya na pinagmumulan at anong mga pamamaraan ng pangangalap ng datos ang gagamitin? *Describe the methodology and methods of the study.*

A. Technical Approach

Fieldwork will be conducted between January to August 2020. A total of two hundred fifty (250) traditional health practitioners from the eastern side of the municipalities of Camarines Sur – Presentacion, Garchitorena, Caramoan, Logonoy *North*, Tinambac, and Siruma (see Figure 1) – will be interviewed separately at different occasions. The study area is a coastal-mountainous community. These municipalities are identified with high incidence of poverty and have relatively poor access to quality healthcare facilities. The inclusion criteria for selecting study participants will be patterned on the ethnobotanical study of Del Fiero and Nolasco (2013), these are (i) has been a folk healer (in any kind) for more than ten years, and (ii) was born and has lived and practicing in the remote east-coastal communities of Camarines Sur for not less than twenty years.



Figure 1. Map showing the study are (inside the blue ink)

The study will use a mixture of quantitative and qualitative methods. Accordingly, the study will use qualitative method for the first three objectives and quantitative for the last objective.

For objective 1 & 2, a directed narrative content analysis will be conducted to gather themes that emerged in the interview text. Narrative qualitative methods offer a vehicle where folk healers can discuss healthcare practices (Riessman (1993). Narrative accounts have been found effective in gathering data from rural persons (Mammen & Sano, 2012; Mentsen, Hellzen, & Enmarker, 2014) and are particularly effective in capturing the experience and reality of folk healers' ethno-medical knowledge and healthcare practices (Creswell, 2014).

Objective 3 will use free-listing, an anthropological interviewing technique used in cultural domain analysis, will be employed to better understand the boundaries of what is being studied, in this case, herbal/medicinal plants and healthcare practices. The use of free-listing is to ensure that the concepts will be defined by the informants themselves, and are, therefore, culturally relevant. The folk healers will be requested to mention as many herbal/medicinal plants as they could think of and as many reasons for identifying these. They were also requested to present samples of these plants. To further explore the data obtained from the free-listing exercise, they will be asked about their perceptions regarding what plant(s) is/are being used to cure certain types of ailments, and which of the plants are frequently used in healing. The herbalists will explain why a plant or a group of plants are important in providing treatment or relief to patients with such illness episodes. Care will be taken to focus on the exact Bikolano (the local language in Camarines Sur) terms used.

Objective 4 will calculate the informant consensus factor (F_{ic}), Use Value (UV), and Fidelity Levels (FL) for each medicinal plant species used to cure various ailments.

Informant consensus factor (F_{ic}) will be used to identify plants of particular intercultural relevance and to agree on their use. The F_{ic} will be used to calculated as the number of use citations in each category (Nur) minus the number of species used (Nt), divided by the number of use citations in each category minus one (Heinrich et al., 1998):

$$F_{ic} = \frac{N_{ur} - Nt}{N_{ur} - 1}$$

F_{ic} values range from 0.00 to 1.00. High F_{ic} values are obtained when only one or a few plant species are reported to be used by a high proportion of informants to treat a particular category, whereas low F_{ic} values indicate that informants disagree over which plant to use.

Fidelity Levels (Fl) is defined as the ratio between the number of informants who independently suggested the use of a species for the same major purpose and the total number of informants who mentioned the plant for any use. In the Fl, Np is the number of informants that reported a use of a plant species to treat a particular disease, and N is the number of informants that used the plants as a medicine to treat any given disease (Friedman et al., 1986);

$$Fl = \frac{Np}{Nx100}$$

The **Use-Value (UV)** index is used to calculate the citation of plants during interviews, proposed by Phillips and Gentry (1993) and adapted by Albuquerque et al. (2007). It is calculated as follows:

$$UVc = \sum U_s/ns$$

where U is the sum of the total number of use citations by all informants for a given species, divided by the total number of informants (ns). This method evaluates the relative importance (RI) of each medicinal species based on its relative use among informants. This index is useful for the analysis of the use of a single species and to compare plants among the same sample.

B. Implementation Approach

In data gathering, the researcher will employ 2-5 enumerators who assist the researcher in identifying the participant-folk healers and in collecting data before, during and after the interview of the researcher. The researcher, himself, will conduct the interview and fieldwork with the assistance of the enumerators in collecting prior information and other follow-up information or data that are needed in the study such as the identification of ethnomedicine and documenting the methods of extraction, preparation and administration that will not be collected during the interview and field work of the researcher. Prior to the conduct of data gathering, they will be given proper instruction and guidelines. The enumerators will be task to identify key informants in each municipality; assist, record, and transcribe KI interviews; and search and document all medicinal plants identified by the KIs.

The interview will be collected through a voice recorder and will be analyzed after the interview. The herbs and other heath practices will be captured through photo or video documentation. Each ethnomedicine such as herbs will be taken picture for documentation and scientific name identification. Books such as List of Philippine Herbal Medicinal Plants of Godofredo Stuart (2019), A dictionary of the Plants Names of the

Philippine Island of Elmer Merrill (1903) and Dictionary of Philippine Plant Names Volumes I and II of Madulid (2001).A sample ethnomedicine will also be filed through the filing folder or clear book with proper labeling.

The proponent will travel to the UPLB Library where the books mentioned above are available. He will also ask assistance from the UPLB Herbarium and expert botanist for the expert identification of the plant species.

To comply with ethical concerns in data collection, only those participants who agreed to be interviewed, voice recorded, and photo documented (with proper written consent) will served as the participants of the study.

6. Ano ang mga target na output ng proyekto? (Kung meron, tulad kung ito ay para sa isang produksyon sa teatro, isang nobela, o video documentary, o mga lesson plan, o iba pa.) *What are the intended outputs? (If any, e.g., a play production, novel, video documentary, lesson plans, others.)*

The following are the expected deliverables of the study:

a. Provide profile and status of traditional health practitioners in the remote east-coastal communities of Camarines Sur and their perception and concept of health and healthcare management. A **coffee table book*** and a **video documentary** will be developed featuring the distinct culture in health and healthcare management;
b. List and compilation of medicinal plants used by traditional health practitioners which can be used for botanical, pharmacological and chemical research;
c. Generate two (2) publications to ACI/ISI/Scopus-indexed and/or JIP participating journals and two (2) presentations to regional, national and/or international research conferences to disseminate the information gathered by the study for **policy making** purposes and to **promote the medicinal culture** of the province to the scientific community and other intended readers.
d. Provide input for possible **culturally sensitive** extension services, projects or programs for the traditional health practitioners and the preservation of culture on traditional health and healthcare management in Camarines Sur.

**Coffee Table Book needs a separate funding for reproduction.*

7. Ilarawan kung ano ang hangarin ng proyekto = ano ang kabuluhan nito o kung paano ito makatutulong; ano ang kontribusyon nito na bagong kaalaman o kaya'y kontribusyon sa sining at kultura ng Filipinas. *Describe intended outcomes—what is the significance of the study or its contribution to human knowledge and to Philippine culture and arts.*

The study will provide information on herbal/medicinal plants and healthcare practices used during treatment episodes, procedures for preparation and administration to achieve the desired results, and illnesses, ailments, and signs and symptoms for which the plants and healthcare practices are used for. The effectiveness and safety of the ethno-medicines which will be explored by this study can be further studied through botanical, pharmacological and chemical research. New medicine or natural medicine can be invented or discovered to cure various illness. Further, this will document the distinct culture of remote communities of Camarines Sur in traditional healthcare management which is facing rapid disappearance.

8. Magbigay ng timetable o iskedyul ng proyekto. *Timetable/Work Plan.*

Activities	*Timeframe*	*Cost/ Fund*	*Sources of Funds*
1. MOA Signing		₱ 5,000.00	NCCA Grant
2. Meeting and briefing with enumerators/ research assistants	1st to 2nd Week of January, 2020	2,500.00	
3. Identification of the key-informants	3rd to 4th Week of January, 2020	5,000.00	
4. Collection of Data	February to August, 2020	235,000.00	
5. Travel to UPLB Main Library and Herbarium & Identification of Plant Species	September, 2020	35,000.00	
6. Analysis of Gathered Data	March to September, 2020	2,500.00	
7. Printing of the Manuscript, Presentation of Research Output, Editing of Video Documentary, and Drafting of Coffee Table Book	October, 2020	15,000.00	

9. Ano ang mga pagkakagastusan at magkano ang kailangan sa bawat isa.*Line Item Budget.* (OR USE PRESCRIBED NCCA FORM)

I. Supplies and Materials	*Particulars*	*Total Amount*	*Source*
Book paper (sub. 20, letter size)	*15 reams*	₱ 3,270.00	NCCA Grant
Book paper (sub. 20, long size)	*10 reams*	2,260.00	
*Epson ink (black)**	*10 bottles*	3,700.00	
*Epson ink (5 each cyan, magenta, yellow)**	*15 bottles*	5,550.00	
*Double-sided photopaper, 160 GSM, A4 size, 50's**	*50 packs*	12,500.00	
*Clear Book Spiral Long***	*250 pcs*	12,500.00	
Ball pen	*10 boxes*	1,100.00	
Notebook	*25 pcs*	375.00	
Printer with Scanner	*1 set*	14,999.00	
*DSLR with lens 24.2 MP****	*1 set*	35,000.00	
*Memory Card (for DSLR) - 64GB****	*1 pc*	4,000.00	
Voice Recorder	*1 pc*	5,000.00	
	Sub-Total	**₱ 100,254.00**	
II. Researchers' Travel Expenses, Representation, & Communication			
Travel Expense (including transportation, meals, accommodation and incidental expenses)		35,000.00	
Representation Expense		10,000.00	
Communication Expense		4,746.00	
	Sub-Total	**₱ 49,746.00**	
III. Enumerators' Fee/Research Assistant Fee			
250 informants x PhP 500.00 each****		**₱ 125,000.00**	
IV. Plant Species Identification Fee (Through the UPLB Herbarium by expert botanists/biologist)		**₱ 25,000.00**	
	Total	**₱ 300,000.00**	

**each plant species will be collected, taken picture and printed for filing and expert identification*
*** each folk healer will have an assigned clear book folder where all medicinal plants/species will be compiled.*
****The study includes capturing the plant species, the actual healthcare practices for the development of coffee table book and video documentary*
*****Enumerator/Research Assistants' Fee includes the identification of key informants; conduct, recording and transcribing of KI interviews; and searching and documentation of all medicinal plants identified by each KI)*

10. Sinu-sino ang magsasagawa ng pag-aaral? Ibigay ang mga pangalan ng Research Team, sino ang Lead Researcher at mga katulong niya, at anong mga gagawin nila? Ilakip ang maikling talambuhay; magbigay ng detalye ng kanilang karanasan sa pananaliksik at kung sila ay nagtatrabaho sa isang institusyon. Ibigay ang mga detalye nito. *Name/s of Lead Researcher and Co-researchers (if any), short bio describing their qualifications and current institutional affiliations (if any) and contact details.*
Mr. Calyd T. Cerio is an Instructor II and the Director of the Center for Bioresource-based Enterprise Development of Partido State University. He earned his Master of Science in Rural Sociology at the University of the Philippines Los Baños and Master of Business Administration at Partido State University. He is an elected member of two (2) prestigious honor societies – Phi Kappa Phi Honor Society and Phi Sigma Biological Sciences Honor Society. He has four (4) published research articles focusing on cultural change, community assessment and sociology of poverty to national and international refereed journals and one (1) research paper accepted for publication in ISI/Scopus-indexed journal. He has also presented research and extension papers to national and international conferences.
11. Kung isang organisasyon ang proponent, ibigay ang detalye ng pangangasiwa ng proyekto. *Research management team.*
N/A

12. Declaration of Ethical Conduct of Research *(*For projects (especially thesis/dissertation) that have already undergone an Ethics Review) **Not Applicable – A signed declaration or commitment statement on ethical conduct of the researcher and Informed Consent Form are submitted.**

Pinatutunayan ko na ang pag-aaral na ito, _____ ay isinalang na sa pagsusuri ukol sa angkop at responsible ng pagsasagawa ng pag-aaral/pananaliksik – *Ethical Conduct of Research* -- ng _____at ang proyekto ay maaari nang isagawa.

I hereby declare that the project submitted herein, _____, has been reviewed for Ethical Conduct of Research by _____ (name of institution) and has been approved for implementation. The following documents are appended (check all that apply):

- o Signed Certification from the Ethics Review Board (should bear a verifiable name of the chair or adviser, the institution, and contact details)
- o Signed Approval/clearance from client groups (e.g., NCIP)
- o Informed Consent / Free Prior and Informed Consent
- o Other: _____

CALYD T. CERIO
SIGNATURE OVER PRINTED NAME OF PROPONENT

Literature Cited:

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